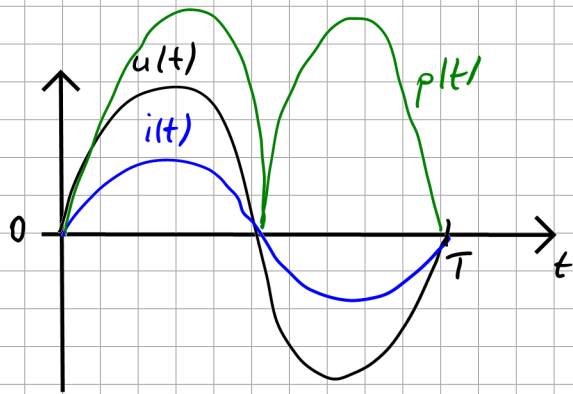
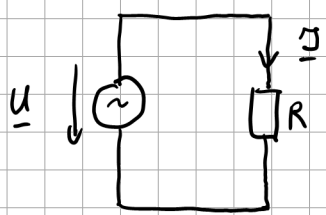


12 Mehrphasensysteme (Drehstromsystem)

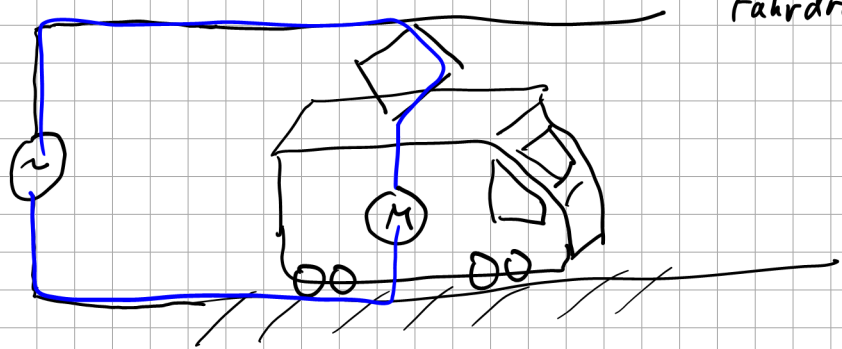


$$p(t) = u(t) \cdot i(t)$$

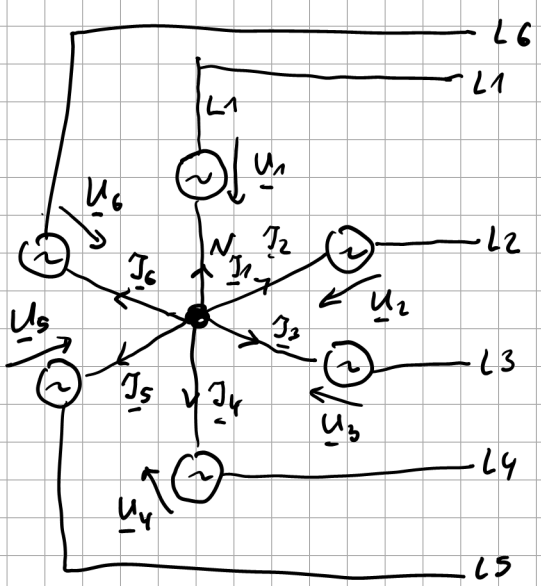
Bsp: Motor: $p_{\text{mech}} = M_d \cdot \omega = p_{\text{el}}(t) = u(t) \cdot i(t)$

$\uparrow$ $\uparrow$
 Drehmoment Kreisfrequenz

Bsp: Einphasenverbraucher: Bahn

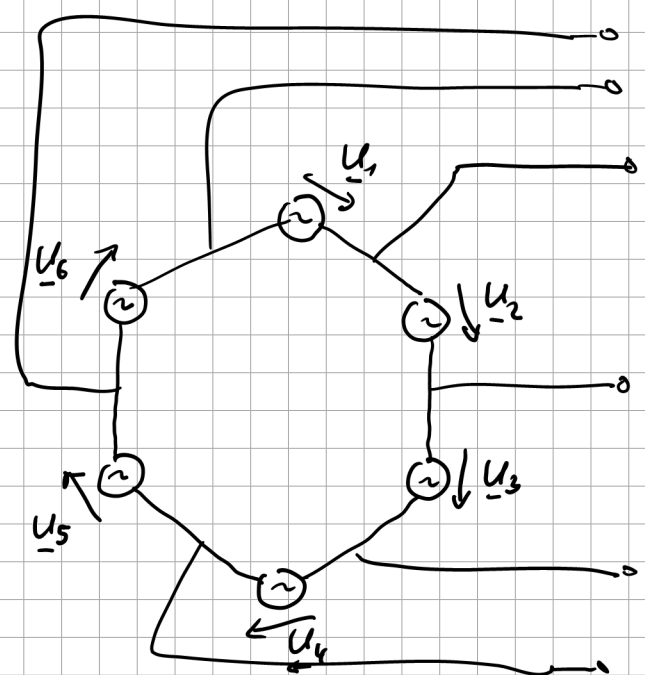


Fahrdrabt: 15 kV_{rms}
16,7 Hz



Sternschaltung:

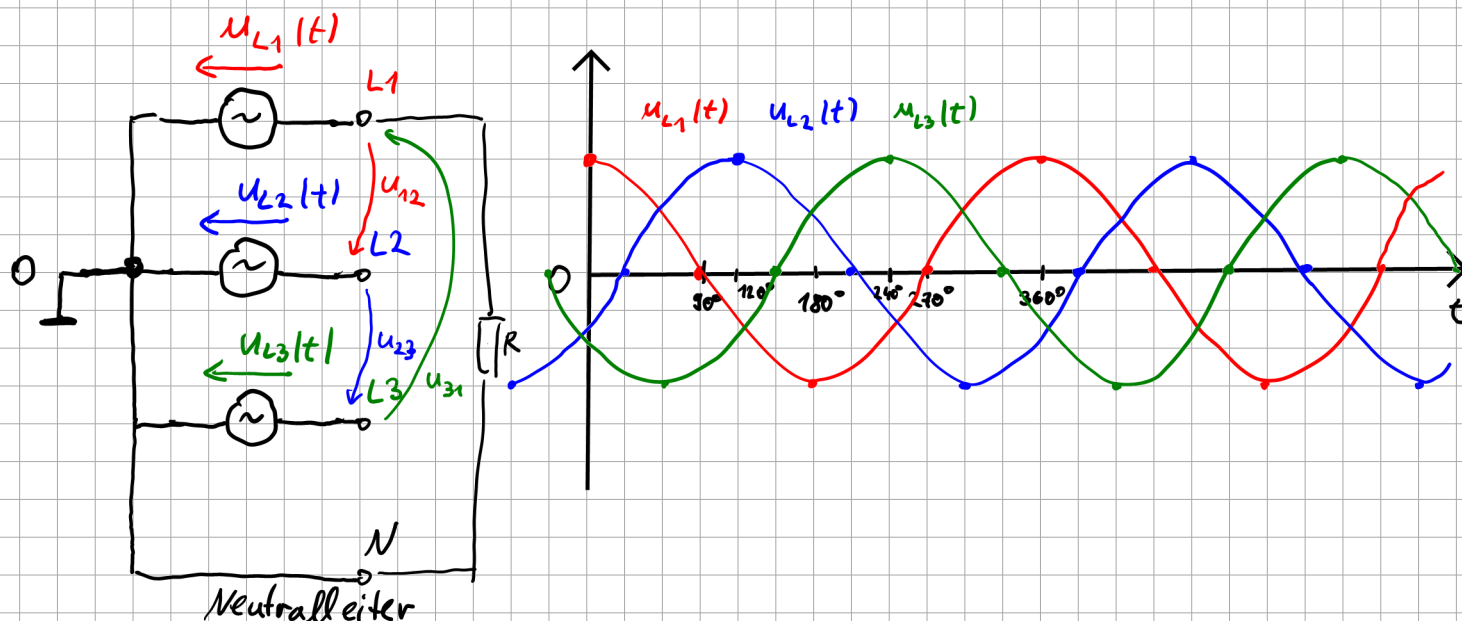
$$\sum_i \underline{I}_i = 0$$



Ringschaltung:

$$\sum \underline{U}_i = 0$$

Symmetrisches Dreiphasensystem in Sternschaltung



$$u_{L1}(t) = \hat{u}_\lambda \cdot \cos(\omega t) = \sqrt{2} U_\lambda \cdot \cos(\omega t)$$

$$u_{L2}(t) = \hat{u}_\lambda \cdot \cos(\omega t - 120^\circ) = \sqrt{2} U_\lambda \cdot \cos(\omega t - 120^\circ)$$

$$u_{L3}(t) = \hat{u}_\lambda \cdot \cos(\omega t - 240^\circ) = \sqrt{2} \cdot U_\lambda \cdot \cos(\omega t - 240^\circ)$$

Symmetrie: Amplituden gleich, $\varphi_{u1} = 0^\circ$, $\varphi_{u2} = -120^\circ$, $\varphi_{u3} = -240^\circ$

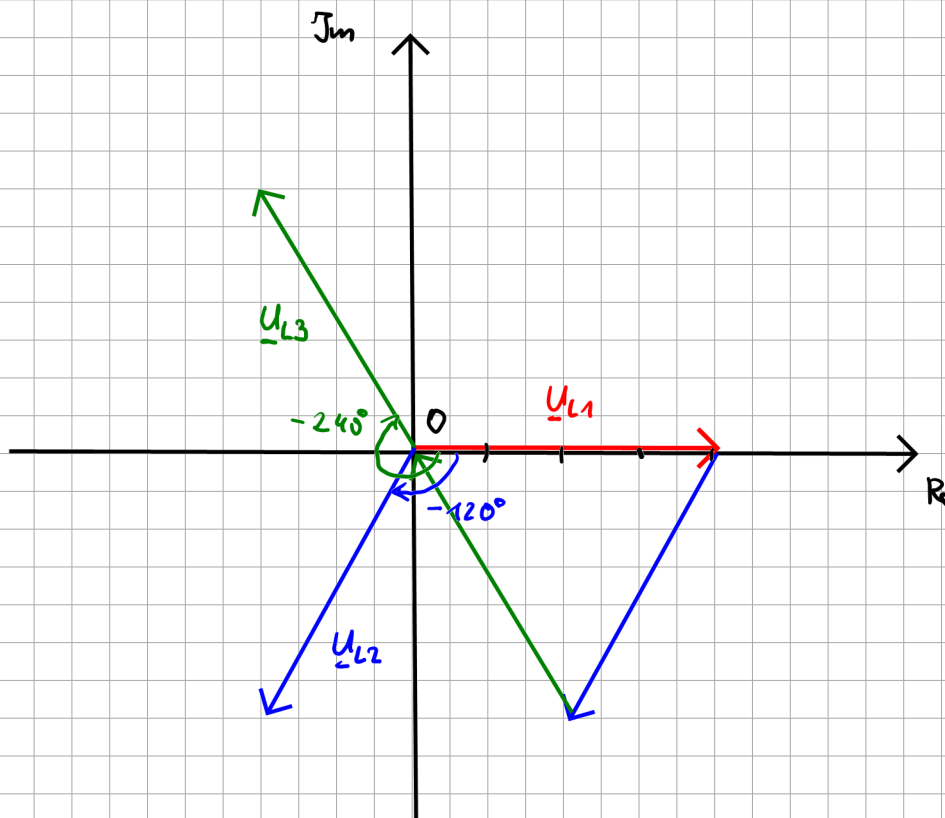
Zeigerdarstellung:

$$e^{jx} = \cos x + j \sin x$$

$$\underline{u}_{L1} = U_\lambda \cdot e^{j0^\circ} = U_\lambda \cdot 1$$

$$\underline{u}_{L2} = U_\lambda \cdot e^{-j120^\circ} = U_\lambda \cdot \left(-\frac{1}{2} - j \frac{1}{2} \sqrt{3} \right)$$

$$\underline{u}_{L3} = U_\lambda \cdot e^{-j240^\circ} = U_\lambda \cdot \left(-\frac{1}{2} + j \frac{1}{2} \sqrt{3} \right)$$



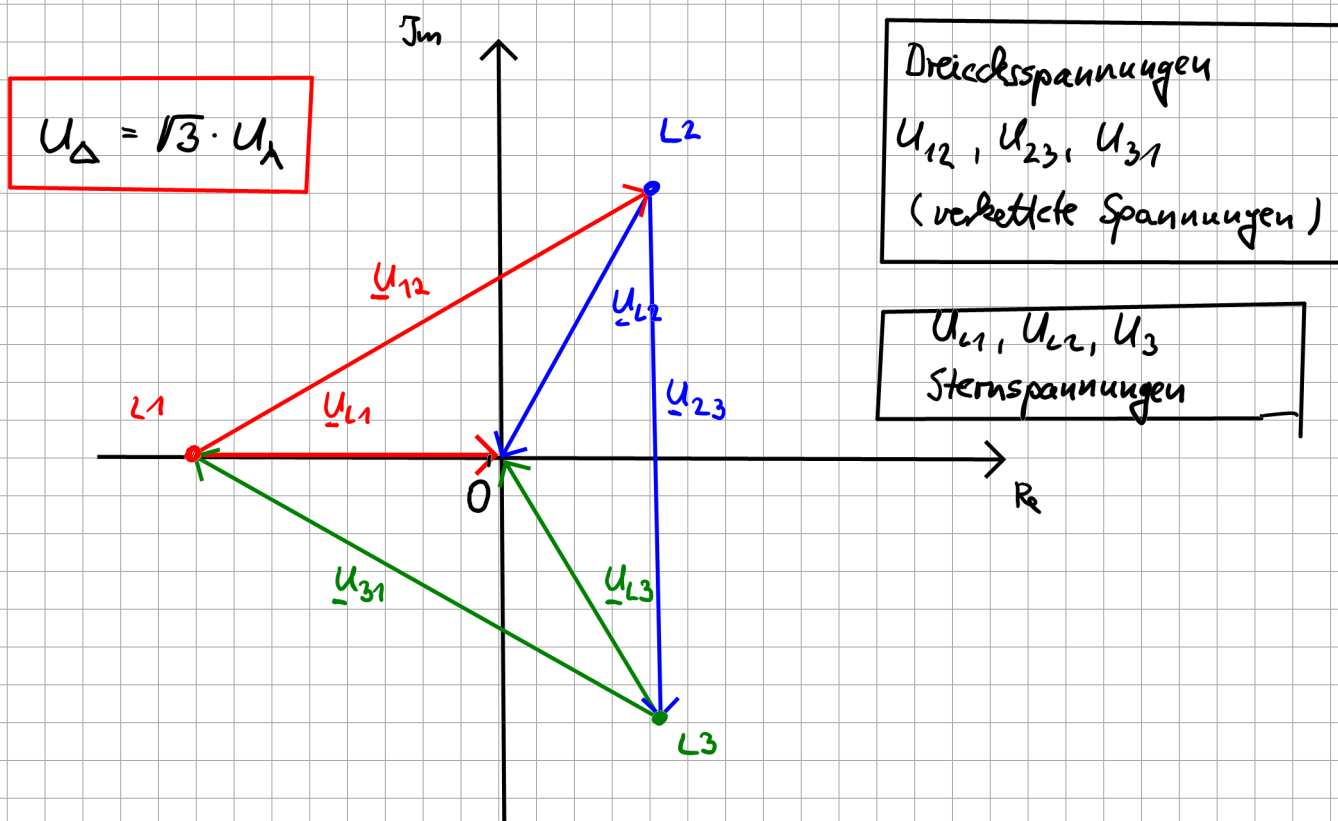
$$\underline{U}_{L1} + \underline{U}_{L2} + \underline{U}_{L3}$$

$$\begin{aligned}
 &= U_\lambda \cdot 1 + U_\lambda \left(-\frac{1}{2} - j\frac{1}{2}\sqrt{3} \right) + U_\lambda \cdot \left(-\frac{1}{2} + j\frac{1}{2}\sqrt{3} \right) \\
 &= U_\lambda \left[1 - \frac{1}{2} - j\frac{1}{2}\sqrt{3} - \frac{1}{2} + j\frac{1}{2}\sqrt{3} \right] \\
 &= 0!
 \end{aligned}$$

and im Zeitbereich .

$$u_{L1}(t) + u_{L2}(t) + u_{L3}(t)$$

$$\begin{aligned}
 &= \sqrt{2} \cdot U_\lambda \left[\underbrace{\cos(\omega t) + \cos(\omega t - 120^\circ) + \cos(\omega t - 240^\circ)}_0 \right] \\
 &= 0
 \end{aligned}$$

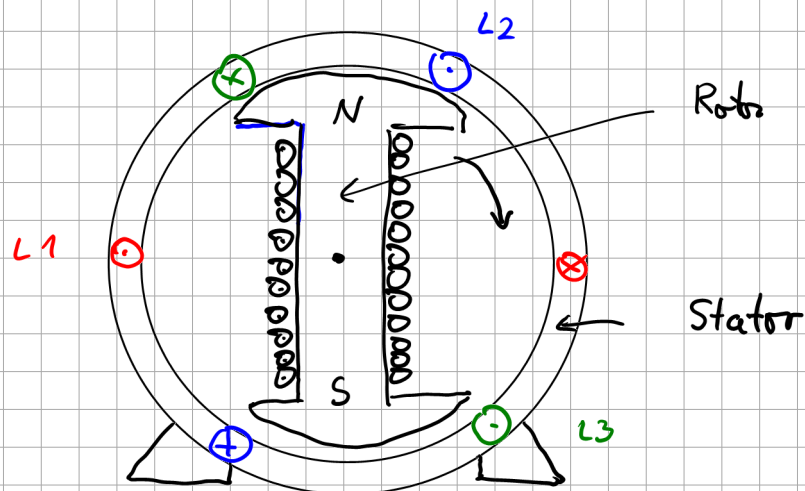


$$\begin{aligned}
 \underline{U}_{12} &= \underline{U}_{L1} - \underline{U}_{L2} \\
 &= U_{\lambda} \cdot 1 - U_{\lambda} \cdot \left(-\frac{1}{2} - j\frac{1}{2}\sqrt{3}\right) \\
 &= U_{\lambda} \cdot \left[1 + \frac{1}{2} + j\frac{1}{2}\sqrt{3}\right] \\
 &= U_{\lambda} \cdot \left(\frac{3}{2} + j\frac{1}{2}\sqrt{3}\right) \\
 &= U_{\lambda} \cdot \sqrt{\left(\frac{3}{2}\right)^2 + \left(\frac{1}{2}\sqrt{3}\right)^2} \cdot e^{j \arctan\left(\frac{\frac{1}{2}\sqrt{3}}{\frac{3}{2}}\right)} \\
 &= U_{\lambda} \cdot \sqrt{\frac{9}{4} + \frac{3}{4}} \cdot e^{j \arctan\left(\frac{1}{\sqrt{3}}\right)} \\
 &= U_{\lambda} \cdot \sqrt{3} \cdot e^{j30^\circ}
 \end{aligned}$$

$$\underline{U}_{12} = \sqrt{3} \cdot U_{\lambda} \cdot e^{j30^\circ}$$

Europa: $U_{\lambda} = 230V$, $U_{\Delta} = 400V$

Spannungserzeugung im Generator



Schenkelpolmaschine

Polpaarzahl $p = 1$

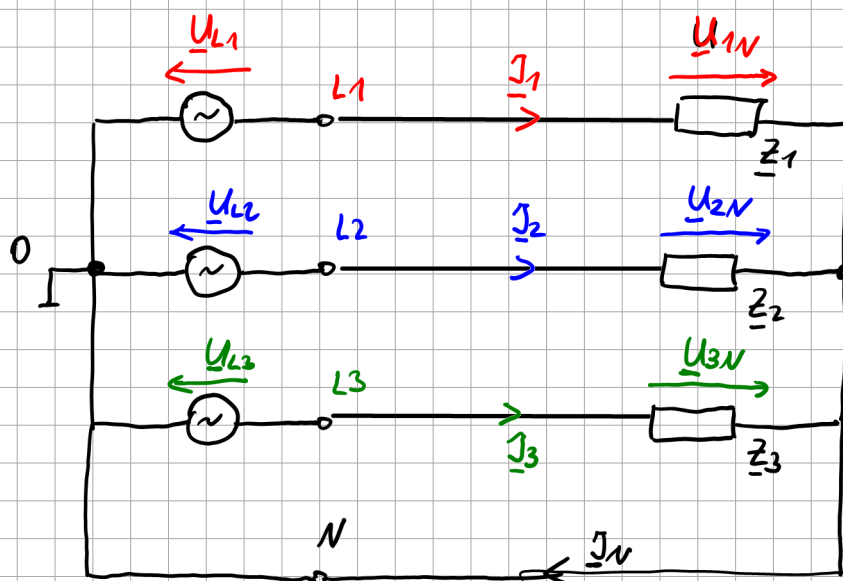
$$f = 50 \text{ Hz} = 50 \cdot \frac{1}{s} = 50 \cdot \frac{1}{\frac{1}{60} \text{ min}} = 3000 \text{ min}^{-1}$$

Turbogenerator : max 1200 MW

$$U_N = 27 \text{ kV}$$

Verbraucher am Drehspannungsnetz

in Sternschaltung, symmetr. Verbraucher: $\underline{Z}_1 = \underline{Z}_2 = \underline{Z}_3 = \underline{Z}$



$$\underline{I}_1 = \frac{\underline{U}_{1N}}{\underline{Z}_1} = \frac{\underline{U}_{L1}}{\underline{Z}_1}$$

$$\underline{I}_2 = \frac{\underline{U}_{2N}}{\underline{Z}_2} = \frac{\underline{U}_{L2}}{\underline{Z}_2}$$

$$\underline{I}_3 = \frac{\underline{U}_{3N}}{\underline{Z}_3} = \frac{\underline{U}_{L3}}{\underline{Z}_3}$$

$$\underline{I}_N = \underline{I}_1 + \underline{I}_2 + \underline{I}_3$$

$$= \frac{1}{\underline{Z}} \left[\underbrace{\underline{U}_{L1} + \underline{U}_{L2} + \underline{U}_{L3}}_{=0!} \right]$$

Satz: beim symmetrischen Drehstromverbraucher im Stern
fließt kein Strom im Neutralleiter $\underline{I}_N = 0!$